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Resilience and academic motivation's mediation effects in nursing students' academic stress and self-directed learning: A multicenter cross-sectional study

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ABSTRACT

Aims: To investigate the mediating role of resilience and academic motivation between academic stress and self-directed learning.
Background: Academic stress in nursing students is a well-reported concept that affects resilience, academic motivation and self-directed learning. However, there is a dearth of studies investigating the mediating role of resilience and academic motivation between academic stress and self-directed learning.
Design: Cross-sectional study and mediation analysis.
Methods: Nursing students (n = 718) were recruited from five nursing schools via convenience sampling. Four self-report scales (Perception of Academic Stress Scale, Connor and Davidson Resilience Scale, Short Academic Motivation Scale and Self-directed Learning Instrument) were used to collect data from August to December 2022. Pearson's r, bivariate analysis and multistage regression analyses were employed to analyze the data.
Results: Academic stress negatively influences nursing students' resilience, academic motivation and self-directed learning. Resilience and academic motivation have a positive impact on self-directed learning. Resilience and academic motivation mediate the relationship between academic stress and self-directed learning, as evidenced by a reduction in the negative impacts of academic stress on nursing students.
Conclusion: Resilience and academic motivation, as mediators, reduce the effects of academic stress on self-directed learning. Nursing educators and administrators should promote programs that strengthen resilience and academic motivation. Thus, improving educational and clinical performance.

1. Introduction

The COVID-19 pandemic emphasized the importance and resilience of nurses. The International Council of Nurses ICN (2021) acknowledged these factors as key to healthcare systems change. Nursing students are the future of the healthcare systems and nursing workforce. However, the pandemic disrupted 73 % of the nursing education programs and 54 % of the post-registration education (ICN, 2021). These findings could complicate the long-standing nursing shortage of nurses worldwide (National League for Nursing, 2021).

In 2021, the American Nurses Association anticipated that more than one million registered nurses would retire by 2030, causing a significant shortage in the nursing industry. The escalating worries of attrition rates and nursing shortages extend to other nations. In other Western and Asian countries, turnover rates for nurses range from 13 % to 44.3 % (ICN, 2019). Similarly, the Philippines experienced the same problems. Although 470 universities in the Philippines offer nursing programs and approximately 80,000 students graduate yearly, the country has shown an exodus of nurse migration in recent years (Dahl et al., 2021).

Nursing students play an essential role in the development of any healthcare system. They remedy the ongoing nursing workforce shortage (ICN, 2021). Nevertheless, nursing students encounter comparable stress-related health difficulties as nurses do (McDermott et al., 2020). Nursing students with low resilience who cannot adapt and manage stress effectively in any academic clinical setting have been recognized as a high-risk group of health workers (Diffley and Duddle, 2022). Hence, compromising future nurses' academic and clinical performance (ICN, 2021).

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2. Background

There is a wealth of literature on nursing students' academic challenges in higher education. Nursing students acquire holistic knowledge, skills and behaviors in nursing school to provide safe, quality care and contribute to positive healthcare outcomes. Due to their demanding workloads, numerous assignments and lack of preparation for clinical practice, nursing students are constantly exposed to stress (Chisholm-Burns et al., 2021). Stress and anxiety from a competitive environment have a negative impact on their academic motivation and self-directed learning (SDL) behaviors (Berdida and Grande, 2023; Wang, 2019). Conversely, nursing students' academic motivation and SDL were reported to have positive correlations (Grande et al., 2022). Undeniably, the stress they face in transitioning into adulthood and acclimating to college life could have a negative impact on their academic success and mental health (Ang et al., 2022).

Stress is regarded as always present throughout the undergraduate course of nursing students. From the standpoint of the academic context, academic stress is defined as a psychological state caused by continual societal and interpersonal pressure that reduces the person's reserves (Trigueros et al., 2020). This state results in students' difficulty adapting to the demands of the academic environment (Bedewy and Gabriel, 2015). Previous studies demonstrated that academic stress was negatively associated with clinical practice–related stress, depression and anxiety among nursing students (Devi et al., 2021; Van Hoek et al., 2019). Similarly, McDermott et al. (2020) reported that nursing students' greater resilience is associated with lower academic stress via the mediation of depression. Because academic stress has a negative impact on students' academic success, it is a common focus of national public health surveys of the general college student population (ICN, 2021).

Resilience is a well-documented protective factor in minimizing the negative impacts of nursing students' stress and anxiety (Ang et al., 2021; Diffley and Duddle, 2022). Resilience is a learning outcome of an individual's experiences that enables individuals to overcome adversity (VanMeter and Cicchetti, 2020). Resilience affects academic performance and nursing students' decision to stay in school. Resilient nursing students report improved academic success, better well-being and decreased turnover, burnout and attrition (Cooper et al., 2021; García-Crespo et al., 2021). Currently, there is an increased interest in developing resilience competency among nursing students. Hence, encouraging nursing educators and policymakers to incorporate resilience competency programs in the nursing curriculum (Berdida and Grande, 2023; Diffley and Duddle, 2022). The mediating effect of resilience on COVID-19 anxiety, academic stress, quality of life, life satisfaction and psychological well-being was reported among Filipino nursing students (Berdida and Grande, 2023; Labrague, 2021). However, the mediating roles of resilience and academic motivation between academic stress and SDL remain unreported.

Another essential topic in nursing education is academic motivation. This type of motivation is either intrinsic or extrinsic (Ryan and Deci, 2000; Vallerand et al., 1992). Intrinsic motivation is engaging in an activity for its own sake and the gratification obtained from it. Conversely, extrinsic motivation is a response to external stimuli, such as the anticipation of rewards or penalties, the chance of promotion at work, or the pursuit of a degree (Ryan and Deci, 2000). Nursing students' academic motivation is multifaceted and influenced by cultural, familial, social, educational and professional factors (Grande et al., 2022). Various factors influence academic motivation, including academic achievement and satisfaction, creativity, stress reduction and preparation for a career as a licensed nurse (Rafii et al., 2019). Academic motivation and resilience could help students achieve high levels of intrinsic motivation and overcome traumatic clinical experiences (Yun et al., 2020). Similarly, a significant negative correlation was found between nursing students' academic stress and academic motivation during nursing skills training (Wang et al., 2019).

SDL is self-initiated learning where one identifies one's learning requirements and goals, seeks learning resources, selects and implements appropriate approaches and assesses one's learning level (Knowles et al., 2014). Academic performance (Cazan and Schiopca, 2014), professional values (Lee et al., 2020), resilience (Zhou et al., 2022) and academic motivation are all connected with SDL (Grande et al., 2022). Many papers have investigated academic motivation in nursing students, and it has been discovered that academic motivation is the most critical factor in enhancing their SDL (Grande et al., 2022; Rafii et al., 2019; Ryan and Deci, 2000). Thus, nurse educators and students are encouraged to identify mechanisms to develop SDL strategies in academic and clinical settings (Zhou et al., 2022). Additionally, SDL abilities allow nurses and allied health workers to sustain lifelong learning (Grande et al., 2022).

Several studies established the relationships between academic stress to clinical practice-related stress (Devi et al., 2021; Van Hoek et al., 2019), resilience (Ang et al., 2021; McDermott et al., 2020), academic achievement (Cooper et al., 2021; García-Crespo et al., 2021) and caring (Zhou et al., 2022); academic motivation to academic achievement (Rafii et al., 2019), intrinsic motivation (Yun et al., 2020) and SDL (Grande et al., 2022); the mediating effect of resilience between COVID-19 anxiety, academic stress and quality of life (Berdida and Grande, 2023); and the mediating effect of learning motivation between stress and academic emotions (Wang et al., 2019). However, studies investigating the interrelationships of academic stress, academic motivation, resilience and SDL among nursing students remain scant. Therefore, this study aimed to investigate how resilience and academic motivation can mediate the adverse effects of academic stress on SDL. Hence, providing theoretical and empirical evidence for enhancing nursing students' resilience, academic motivation and SDL while reducing academic stress.

Mediation analysis explores the relationship between an independent (i.e., academic stress) and dependent variable (i.e., SDL) is mediated by the partial or total effect or facilitation of a third variable, the mediator (i.e., resilience and academic motivation) (Byrne, 2013). In its most basic form, the mediator bridges or facilitates the relationship from the independent variable to the dependent variable, either wholly or partially, which may change the strength of the relationship between the two variables.

2.1. Theoretical underpinning

Two theories were used in this study: Stephens (2013) nursing student resilience model and Ryan and Deci's (2000) self-determination theory. Stephens (2013) nursing student resilience model postulated that resilience results from enhancing protective factors or resources to strengthen coping and adaptive skills when confronted with adversity or stress. Positive emotions, adaptability, competence, social support, self-efficacy, faith, hope, connectivity, self-directed learning, efficient coping, sense of humor, awareness of healthy habits and hazards, perseverance and motivation are examples of protective factors (Stephens, 2013). These factors capitalize on a person's innate capacity for resilience and their demands and abilities. Nurse educators are the best people to offer educational interventions to improve students' protective factors or resources. Building resilience techniques during nursing students' training and practice will help them recognize and improve their protective factors through personal development, effective coping and well-being (Cooper et al., 2021; García-Crespo et al., 2021; Stephens, 2013). As a protective factor, nursing students' resilience decreases academic stress, depression and anxiety (Ang et al., 2021; McDermott et al., 2020). Additionally, resilience mediates the impact of academic stress and anxiety on quality of life Berdida and Grande, 2023). Exploring nursing students' perceptions and development of resilience can provide insights for future work and inform institutional policy, as resilience can potentially mitigate the effects of stress (i.e., academic stress).

Self-determination theory (Ryan and Deci, 2000) provided a lens for examining academic motivation, academic stress and self-directed learning. The theory's three categories of motivation, namely intrinsic and extrinsic motivation and amotivation, represent differing degrees of self-determination (Ryan and Deci, 2000). Intrinsic motivation is the desire to engage in enjoyable and pleasurable things. Extrinsic motivation, conversely, is the proclivity to perform tasks for unrelated reasons, such as anticipated remuneration or penalty. Amotivation is not acting at all or doing without intent due to feelings of ineptitude, not expecting the intended outcome, or undervaluing an activity (Ryan and Deci, 2000). Recent studies reported that academic motivation has a positive impact on self-directed learning among nursing students (Grande et al., 2022), while stress negatively correlates with academic motivation (Wang et al., 2019). Moreover, nursing students' increased resilience mediated traumatic clinical experiences' negative impacts on academic motivation (Yun et al., 2020).

Therefore, strengthening resilience and academic motivation as protective factors underpin the nurse educator strategies to promote nursing students' SDL while mitigating academic stress. Resilience and academic motivation must be incorporated into the nursing curriculum to support students in overcoming obstacles and provide structured venues to learn how to succeed in their academic and clinical practice environments (Hughes et al., 2021).

2.2. Research hypothesis

Considering the prior studies, this study hypothesized that (a) academic stress negatively influences nursing students' resilience, academic motivation and SDL (Hypothesis 1); (b) resilience and academic motivation positively influence SDL (Hypothesis 2) and (c) academic stress influences nursing students' SDL via the mediating effect of resilience and academic motivation (Hypothesis 3).

3. Methods

3.1. Design

A correlational cross-sectional study design followed the STROBE guidelines (Supplementary file 1). Also, the mediation model was constructed to display the significant relationships among study variables.

3.2. Sampling, participants, setting

To recruit participants, convenience sampling was used. Participants must be officially enrolled nursing students who have completed at least one semester in one of the five research sites. 1000 Google survey forms were disseminated, and 787 completed forms were collected. Only 718 forms (a response rate of 71.8%) were determined acceptable for data analysis after data cleaning. The sample size for the mediation analysis was calculated by multiplying the total number of scale items used by ten ([N = 10 +14 +20 +18]*10 =620) (Huang et al., 2017; Sim et al., 2022). Excluding the 35 participants in the pilot test, the total number of participants met the sample size requirement for the mediation analysis.

Participants in the study were selected from five nursing schools in the Philippines (3 public and two private). The first nursing college is part of a local public institution in Manila and established its nursing programs in the 1990s. The second study location is a public research university in the southern portion of Luzon Island. It was established in the late 1960s and the college in the 1990s. The third nursing school is a constituent of a public state university in Samar Island. The two private institutions were nursing schools in Luzon Island's northern and southern provinces. Overall, there were 1950 eligible participants in these study settings.

3.3. Ethical considerations

The ethics review board of a local institution in Manila approved the implementation of this study (Protocol No.: XXX-ERC-0024, approved: 05/17/2022). Data gathering commenced after all the ethical requirements had been complied with. The first section of the online Google survey forms contained information on the study and the consent statement. English was used to administer the survey instruments. Participants impliedly agreed to participate in this study when they filled out and returned the survey form. Additionally, participant anonymity and data confidentiality were upheld throughout the study while adhering to the Helsinki Declaration (2013). The collected data from the participants will be deleted two years after publication.

3.4. Instruments

The instrument used in this study had two sections. Section 1 obtained participants' demographic profiles (e.g., age, gender, year level, general weighted average [GWA], type of class attended [hybrid or full online], religion and experienced failure in a course). The year level refers to the academic level classification (i.e., the first year means the first academic year or freshman year). In the Philippines, GWA is used to grade a college or university student's academic performance. It is the average of students' marks across all courses they have taken in a semester or academic year. There is a corresponding label in every GWA range (e.g., 89–94%: superior).

Section 2 contained the four standardized scales: the 18-item Perception of Academic Stress Scale (PASS; Bedewy and Gabriel, 2015), the 10-item Connor and Davidson Resilience Scale (CD-RISC; Campbell-Sills and Stein, 2007), the 14-item Short Academic Motivation Scale (SAMS; Kotera et al., 2021), the 20-item Self-directed Learning Instrument (SDLI; Cheng et al., 2010). This study sought permission from the original authors to use the scales.

3.4.1. Academic stress

The 18-item PASS was employed to evaluate academic stress. The scale is rated on a five-point Likert scale ranging from 1 (highly disagree) to 5 (strongly disagree). Five items of the scale were reverse-scored to prevent response pattern errors (e.g., "I am confident that I will be a successful student," "The time allocated to classes and academic work is enough"). The scale has a score range of 5–90. A reduced score indicates more academic stress. Cronbach's alpha for the PASS scale is 0.70 (Bedewy and Gabriel, 2015). It was 0.91 in this study. This instrument has been used and validated among Australian (Brown et al., 2018), Korean (An et al., 2022; Park et al., 2022) and Filipino (Berdida and Grande, 2023) nursing students.

3.4.2. Resilience

The 10-item CD-RISC "measures the attainment of positive functioning of a person in the face of adversity" (Campbell-Sills and Stein, 2007, p. 1020). This scale contains questions such as "Able to adapt to change" and "Tend to bounce back after illness or hardship." Each item is scored on a five-point scale (1 = "not true at all" to 5 = "almost always true"). The range score is 5–50. A higher score means higher resilience. The original Cronbach's alpha of this scale was 0.85 (Campbell-Sills and Stein, 2007). In this study, Cronbach's alpha was 0.97.

3.4.3. Academic motivation

The 14-item SAMS comprises 7 factors: Intrinsic Motivation to Know (e.g., For the pleasure I experience when I discover new things never seen before), Intrinsic Motivation toward Accomplishment (e.g., For the pleasure that I experience while I am surpassing myself in one of my personal accomplishments), Intrinsic Motivation to Experience Stimulation (e.g., For the pleasure that I experience when I read interesting authors), Identified Regulation (e.g., Because I think that a college education will help me better prepare for the career I have chosen), Introjected Regulation (e.g., Because of the fact that when I succeed in college I feel important), External Regulation (e.g., To obtain a more prestigious job later on) and Amotivation (e.g., I can't see why I go to college and frankly, I couldn't care less). Each factor has two items. This scale is scored on a 7-point Likert scale (1 = "does not correspond at all" to 7 = "corresponds exactly"). This instrument yields a range score of 7–98. A higher score indicates higher academic motivation. The Cronbach's alpha among the seven factors ranges between 0.63 and 0.86, indicating acceptable to high reliability (Kotera et al., 2021). This scale has been used among British healthcare students (Kotera et al., 2021). The overall Cronbach's alpha for this study was 0.93.

3.4.4. Self-directed learning

The 20-item SDLI was developed and validated among nursing students (Cheng et al., 2010). SLS consists of four domains: Learning motivation (6 items), Planning and implementing (6 items), Self-monitoring (4 items) and Interpersonal communications (4 items). All 20 items are positively worded. Examples of item questions include "I know what I need to learn" and "Regardless of the results or effectiveness of my learning, I still like learning." Each item is rated on a 5-point Likert scale (1 for "strongly disagree'' to 5 for "strongly agree''). The score range of SDLI is 20–100. Thus, a higher score signifies a higher SDL. Cronbach's alpha was 0.916 for the entire instrument and 0.801 (Learning motivation), 0.861 (Planning and implementing), 0.785 (Self-monitoring) and 0.765 (Interpersonal communications) for the four domains (Cheng et al., 2010). This study showed an overall Cronbach's alpha of 0.95.

3.5. Validity and rigor

The scales were pilot tested (n = 35) before data collection. Participants in the pilot test were excluded from the final study sample. The original English versions were used in this study since the participants were English proficient. The pilot test aimed to assess whether there were any issues with the scale interpretation and whether any items were regarded as unsuitable for Filipino nursing students. Good to excellent reliability coefficients were obtained after reliability analyses, with Cronbach's alpha values were 0.97, 0.91, 0.93 and 0.95 for the CD-RISC, PASS, SAMS and SDLI, respectively. These results support the validity and reliability of the study instruments (Taber, 2018).

3.6. Data collection

Data collection was conducted from August 22 to December 5, 2022. Google forms were used to disseminate the survey instruments because the COVID-19 safety protocols prohibited face-to-face data collection. The college deans of the four study locations and one university's ethics and research center approved the distribution and collection of the survey forms. Point persons (e.g., clinical placement coordinators or department heads) were identified and the survey link was provided to them. The point persons forwarded the survey link to their students' emails and social media group chats (e.g., Messenger, WhatsApp). The participants can access the survey link using a laptop or a smartphone. The survey form cannot be submitted unless all questions have been addressed. Accordingly, participants can withdraw from the study whenever they want. The survey can be accomplished within 20–25 min. Once participants click the submit button in the survey form, the data will be saved in the Google Drive repository. This password-protected Google Drive was only accessible to the researcher. The final data analysis did not include survey forms that were completed in less than 15 min. They were likely answered without reading the items on the instruments.

3.7. Data analysis

Participants' demographic profiles and responses to the scales were computed using means, standard deviations and percentages. Pearson's correlation coefficient (*r*) and bivariate analysis were used to determine the correlations between the key study variables. Multistage regression analyses were employed to determine the mediating effect of resilience and academic motivation on the relationship between academic stress and SDL. First, the independent variable (academic stress) was regressed on the mediating variables (resilience and academic motivation). Second, the mediating variables were regressed on the dependent variable (SDL). Third, the mediating variables were regressed on the dependent variable while controlling for the independent variable. The Sobel test validated the significance of mediation analysis.

The Kolmogorov–Smirnov test was used to determine the data's normality. Variables that were missing were neither replaced nor estimated. However, there were no missing data. The variance inflation factor for the various predictor variables was within acceptable limits (<10). This implies that multicollinearity was minimal or nonexistent. The Durbin–Watson test yielded a result close to 2, confirming the absence of autocorrelation. The graph of the residual values and the expected values revealed homoscedasticity.

4. Results

4.1. Participants' demographic profiles

A total of 718 participants were included in the study (response rate: 71.8 %). Table 1 shows the distribution of participants according to demographic profiles. Their age ranged from 18 to 24 years, with a mean of 20.73 years (SD = 1.75 years). There were more females (82.1 %), with a male-to-female ratio of 1:5. More than half of the participants were third (23.7 %) and fourth year (29.1 %) students. Most participants' GWA ranged from 89 % to 94 % (69.1 %).

4.2. Correlations between academic stress, resilience, academic motivation and SDL

This study obtained the following overall mean scores of study variables: resilience = 4.16 (SD=0.31), academic stress = 3.83 (SD=0.31), SDL = 4.11 (SD=0.39) and academic motivation = 5.17 (SD=0.52). Academic stress was negatively correlated with resilience (r = −0.122, *p* = 0.002), academic motivation (r = −0.457, *p* < 0.001) and SDL (r = −0.118, *p* = 0.002). Resilience was positively correlated to SDL (r =

Table 1
Participants' demographic profile (n = 718).

Profile variables	Frequency	Percentage
Age (in years)		
Mean = 20.73 (SD=1.75)		
Gender	129	17.9
Male	589	82.1
Female		
Year Level	231	32.2
First Year	108	15.0
Second Year	170	23.7
Third Year	209	29.1
Fourth year		
General Weighted Average (GWA)	52	7.3
Excellent (95–100 %)	496	69.1
Superior (89–94 %)	170	23.6
Very Good (83–88 %)	0	0
Above Average (77–82 %)	0	0
Average (<76 %)		
Type of Class Attended	231	32.2
Full online class (attended all courses in an online learning mode)	487	67.8
Hybrid (attended courses both on-campus/hospital setting)		
Religion	625	87.1
Catholic	93	12.9
Non-Catholic		
Experienced failure in a course of subject/s	51	7.1
Yes	667	92.9
No		

0.113, $p = 0.004$) and academic motivation ($r = 0.157$, $p < 0.001$). Similarly, academic motivation was directly correlated to SDL ($r = 0.074$, $p = 0.047$) (Table 2).

4.3. *Mediating effects of resilience and academic motivation on the relationship between academic stress and SDL*

Multi-stage multiple regression analyses were conducted to assess the mediating role of resilience and academic motivation in the relationship between academic stress and SDL (Table 3; Fig. 1). Academic stress demonstrated a significant negative association with resilience ($\beta = -0.122$, $p = 0.002$, 95%CI: −0.205 to −0.047), academic motivation ($\beta = -0.457$, $p < 0.001$, 95%CI: −0.964 to −0.714) and SDL ($\beta = -0.118$, $p = 0.002$, 95%CI: −0.239 to −0.052). Thus, supporting Hypothesis 1. The mediating variables, resilience ($\beta = 0.113$, $p = 0.004$, 95%CI: 0.044–0.224) and academic motivation ($\beta = 0.157$, $p = <0.001$, 95%CI: 0.009–0.098), were positively correlated to SDL (Hypothesis 2).

Correspondingly, the effect of academic stress on SDL ($\beta = 0.104$, $p = 0.\ 008$, 95%CI: 0.010–0.067) was evident when resilience was added to the model, suggesting that resilience is a significant mediating variable in the association between academic stress and SDL. Concurrently, the effect of academic stress on SDL ($\beta = -0.039$, $p = 0.014$, 95% CI: −0.043 to −0.015) diminished when academic motivation was added to the model, suggesting a significant mediating role for this variable. These results confirmed Hypothesis 3.

The Sobel test confirmed the significance of the mediating effect of resilience (Sobel = −2.06, $p = 0.039$) and academic motivation (Sobel = −2.38, $p = 0.017$) on the relationship between academic stress and SDL. In other words, resilience and academic motivation decreased the impact of academic stress on SDL.

5. Discussion

Academic stress among nursing students is a significant factor that affects resilience, academic motivation and SDL. However, studies are scarce investigating the mediating role of resilience and academic motivation between academic stress and SDL. Hence, this study tested a model that demonstrated the interrelationships among the study variables and the mediating effect of resilience and academic motivation between academic stress and SDL. The following are the study's significant findings. First, academic stress negatively influences nursing students' resilience, academic motivation and SDL. Second, resilience and academic motivation have a direct impact on SDL. Third, resilience and academic motivation mediate the relationship between academic stress and SDL.

5.1. *Hypothesis 1: Academic stress negatively influences nursing students' resilience, academic motivation and SDL*

This study found that academic stress negatively affects nursing students' resilience, academic motivation and SDL. Thus, confirming Hypothesis 1. These findings imply that academic stress has an adverse impact on nursing students' ability to cope with the demands of nursing school. Consequently, impairing nursing students' resilience, academic motivation and SDL. This study was conducted as nursing students were transitioning to the hybrid learning mode (i.e., a combination of online and on-campus classes) due to the COVID-19 pandemic. This could have caused academic stress for the participants. Therefore, in the educational context, decreasing students' experience of academic stress is crucial to improve resilience, academic motivation and SDL. Hence, nursing students may be protected from the negative impacts of academic stress.

It is a known factor that academic stress is the most significant challenge nursing students worldwide experience that affects their academic and clinical productivity (Chisholm-Burns et al., 2021; ICN, 2021; McDermott et al., 2020). Previous studies supported Hypothesis 1 (Berdida and Grande, 2023; Rossi et al., 2021; Wang et al., 2019). Consistent with the findings, Filipino nursing students during the COVID-19 pandemic reported that academic stress had a negative impact on their resilience (Berdida and Grande, 2023). Moreover, Korean student nurses reported a negative correlation between academic stress and motivation during their skills training (Wang et al., 2019). Likewise, nursing students in this study experienced decreased resilience and academic motivation because of their academic stress.

Similarly, nursing students' stress indirectly affected SDL and classroom engagement during a curricular revision (Rossi et al., 2021). At the same time, resilient nursing students manifest lower academic stress (McDermott et al., 2020). Increasing students' resilience and academic motivation have been shown to favorably affect their stress levels and mental health. Students' ability to maintain a positive mental attitude while studying is crucial in developing competence in nursing students and nurses (Wang et al., 2019). The study's findings underscore the importance of stress reduction and self-care programs for students. Nursing school administrators and nurse educators could reduce academic stress by restructuring and decongesting the curriculum.

5.2. *Hypothesis 2: Resilience and academic motivation positively influence SDL*

The participants revealed that increased resilience and academic motivation enhance SDL abilities. Furthermore, resilience directly influences academic motivation. These findings support Hypothesis 2, as evidenced by their positive relationships. Because of the impacts of the COVID-19 pandemic on nursing students' academic lives, it is very conceivable that they have internalized resilience, academic motivation and SDL. It is plausible that nursing students with greater resilience and academic motivation are more likely to be aware of problems and to create a growth mindset to adjust to a challenging environment, which builds better SDL behaviors. Nursing students' resilience could be a critical factor in sustaining academic motivation, thereby enhancing their academic and clinical outcomes. These positive interrelationships of the study variables are crucial not only during a pandemic but also during their academic life and future professional careers.

Notably, a subset of SDLI measured participants learning motivation while SAMS ascertained academic motivation. Learning motivation is theoretically underpinned in Knowles' (1975) self-directed learning theory. Learning motivation is "the inner drive of the learner as well as external stimuli that drive the desire to learn and to take responsibility for one's learning." (Cheng et al., 2010, p. 1155). On the other hand, academic motivation is primarily anchored on Ryan and Deci's (2000) self-determination theory, postulating that human beings' innate ability is to self-actualize in the social community. This theory categorized motivation into intrinsic, extrinsic, or amotivation (Ryan & Deci, 2000). Hence, academic motivation drives students' behaviors critical for academic success (Kotera et al., 2021).

Although there may be a connection between learning motivation and academic motivation, the divergent theoretical basis and the uniqueness of the items of these scales made the distinction between

Table 2
Correlation between key variables.

Variables	Mean ± SD	1	2	3	4
Resilience	4.16 ± 0.31	1			
Academic stress	3.83 ± 0.31	-0.122**, [a]	1		
Self-directed learning	4.11 ± 0.39	0.113**, [a]	-0.118**	1	
Academic motivation	5.17 ± 0.52	0.157**, [a]	-0.457**, [b]	0.074*, [a]	1

*Correlation is significant at the 0.05 level (2-tailed); **Correlation is significant at the 0.01 level (2-tailed)
Coefficient value strength of association (Cohen, 1988): [a]0.1 < | r | < 0.3 Small correlation; [b]0.3 < | r | < 0.5 Medium/moderate correlation; [c]| r | > 0.5 Large/strong correlation

Table 3
Mediating effects of resilience and academic motivation on the relationship between academic stress and SDL.

Structural paths	B	SE	β	t	p	95% CI Lower bound	Upper bound	R^2
Step 1								
Academic Stress →Resilience	-0.126	0.040	-0.122	-3.142	0.002	-0.205	-0.047	0.015
Academic Stress →Academic motivation	-0.839	0.064	-0.457	-13.137	< 0.001	-0.964	-0.714	0.209
Academic Stress →Self-directed learning	-0.145	0.048	-0.118	-3.046	0.002	-0.239	-0.052	0.014
Step 2								
Resilience →Self-directed learning	0.134	0.046	0.113	2.917	0.004	0.044	0.224	0.013
Resilience → Academic motivation	0.277	0.068	0.157	4.062	< 0.001	0.143	0.411	0.025
Academic motivation →Self-directed learning	0.047	0.026	0.074	1.986	0.047	0.009	0.098	0.005
Step 3								
Academic Stress →Resilience →Self-directed learning	0.038	0.014	0.104	2.695	0.008	0.010	0.067	0.011
Academic Stress →Academic motivation →Self-directed learning	-0.029	0.012	-0.039	-2.417	0.014	-0.043	-0.015	0.010

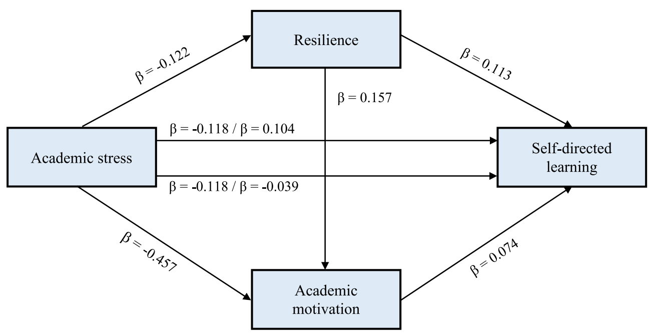

Fig. 1. Final model.

these two concepts. Hence, it is established in this study that these concepts are distinct, but they are interrelated. As such, the study's findings reported that academic motivation directly influences SDL and mediates the negative impact of academic stress on SDL.

Similar to the study's findings, resilience and SDL positively correlated among Chinese nursing students (Zhou et al., 2022). They found that resilience helps students develop SDL skills in acquiring complex knowledge, skill and behaviors during academic and clinical situations. Accordingly, academic motivation and resilience showed positive effects on helping students develop intrinsic motivation and surmount stressful academic or clinical experiences (Yun et al., 2020). Intrinsic motivation was noted as the most significant factor for SDL readiness among Filipino, Thai and Saudi nursing students (Grande et al., 2022). Intrinsic motivation assists nursing students in identifying personal limitations, seeking help from social networks and selecting appropriate solutions to challenges. Therefore, nurse educators should provide curricular and pedagogical innovations to students with lower resilience and academic motivation to increase their SDL knowledge and skills to address the ever-changing academic and clinical challenges.

5.3. Hypothesis 3: Academic stress influences nursing students' SDL via the mediating effect of resilience and academic motivation

This study contributed to the existing body of knowledge by using mediation analysis to explicate the relationships between academic stress, resilience, academic motivation and SDL. The study showed that resilience was a mediator between academic stress and SDL (substantiating Hypothesis 3). This finding indicates that resilience may mitigate the effects of academic stress on students' SDL. Thus, increasing understanding of the potential role of resilience in strengthening SDL amidst academic stress experienced by nursing students.

Additionally, the study's findings underscore the importance of resilience for nursing students to deal with and manage various stressors effectively (Diffley and Duddle, 2022; Devi et al., 2021). Studies conducted before the pandemic (Ang et al., 2021; McDermott et al., 2020) and during the pandemic (Ang et al., 2022; Berdida and Grande, 2023; Labrague, 2021) demonstrated that sufficient resilience and continuous use of positive coping skills not only decreased stress levels in nursing students but also improved overall well-being. Conversely, several research involving nursing students connected a lack of resilience during stressful events to severe physical, mental and social health issues (Yun et al., 2020; Van Hoek et al., 2019). In addition, decreased resilience and less social support were linked to acute stress disorder in students in higher education (Ang et al., 2021). Relative to these findings, it is clear that student nurses must develop resilience behaviors through online or onsite psychoeducation programs.

The mediation analysis showed that academic motivation mediated the impacts of academic stress on SDL. Thus, confirming Hypothesis 3. This finding indicates that nursing students' SDL is maintained due to their academic motivation despite the harmful threats of academic stress. Academic motivation demonstrated a mediating role between nursing students' stress and negative emotions (e.g., loss of control, nervousness and lack of interest) (Wang et al., 2019). They found that academic motivation reduced stress has an impact on student nurses' negative emotions resulting in reduced stress, improved clinical skills and enhanced competence. Academic motivation is a critical component of SDL (Ryan and Deci, 2000; Vallerand et al., 1992). For instance, academic and intrinsic motivation have a direct impact on the SDL readiness of nursing students (Grande et al., 2022). Students who are motivated are likely to engage in independent learning. This characteristic is vital in higher education because most academic and clinical activities require a certain degree of independence (Grande et al., 2022). Although SDL has been proven a successful learning strategy for nursing students and practicing nurses (Lee et al., 2020), not all will develop this characteristic.

Therefore, cognizant of the study's empirical evidence of academic motivation's benefits, nurse educators should explore incorporating innovative and technology-driven teaching and learning approaches that enhance motivation, reduce academic stress and improve SDL. Once nursing students acquire SDL skills during their academic years, they will continue to nurture this learning style beyond graduation. Thus, allowing them to develop lifelong knowledge and professional competence.

By and large, the study was conducted when most Filipino nursing students were adjusting to a hybrid mode of learning curriculum. The hybrid mode of learning entails both online and onsite modes of learning. Theoretical courses were held online, while clinical or skills-related courses were conducted onsite or in hospitals. This curriculum revision was in response to the impacts of the COVID-19 pandemic on the Philippine education system. Starting with the second term of the academic year 2022–2023, higher education institutions (HEIs) must discontinue full online instruction in favor of holding at least half of their classes on campus (Commission on Higher Education [CHED] Memo Order No. 16, 2022). The CHED (2022) required HEIs to adopt hybrid learning and devote at least 50 % of all instructional hours to on-site learning experiences. This change in the delivery of the nursing curriculum could have an impact on nursing students' academic stress, motivation, resilience and SDL. In turn, it may influence their academic outcomes and clinical performance.

5.4. Implications for nursing education

This study has three implications: policy, institutional and course levels. First, at the policy level, the nursing curriculum should include programs (e.g., psychoeducation programs and resilient-skills-enhancing activities) to develop nursing students' resilience and academic motivation. As a result, this will enhance their SDL skills, academic and clinical performance and professional competence. Second, at the institutional level, nursing educators should be trained in implementing and monitoring resilience and academic motivation programs. Moreover, nursing schools should provide a supportive campus environment to help students nurture their resilient characteristics. Finally, at the course level, nursing students who respond negatively to academic stress (e.g., lack of engagement, poor performance) should be provided with early interventions to promote resilience, motivation and SDL. Therefore, improving their holistic well-being and minimizing attrition.

5.5. Limitations and recommendations

There are several limitations encountered in this study. First, four self-report scales were used to collect data. Using these types of scales may result in response and social desirability bias. Second, the hypothesized mediation model was tested using a cross-sectional design that cannot infer the key variables' causality. Third, although this is a multisite study, participants from five nursing institutions were selected via convenience sampling. The sample cannot provide robust representativeness of the entire nursing student population. Thus, the findings should be interpreted cautiously because of the study's limited generalizability.

Based on the study's limitations, the following are recommended. Using a larger sample while recruiting participants through a probability sampling technique. Longitudinal or experimental designs could ascertain the causal relationships between variables over time-sensitive situations. Qualitative research approaches may capture the nursing students' in-depth experiences of the study's variables. Finally, future research could test the proposed model while involving other factors (e.g., social support, assimilation, competence, academic burnout) that may influence the study's variables.

6. Conclusion

This study presented a model of the interrelationships among academic stress, resilience, academic motivation and SDL among nursing students. Academic stress has a negative impact on resilience, academic motivation and SDL. Whereas resilience and academic motivation positively influence SDL. Notably, resilience and academic motivation demonstrated mediating effect between academic stress and SDL. The presented mediation model highlights three crucial directions. First, nursing students' resilience should be strengthened through online or in-person psychoeducational programs that foster resilient behaviors. Second, nurse educators may use innovative, technology-driven teaching and learning strategies that boost motivation, lessen academic stress and increase SDL. Finally, nursing schools should create a supportive campus milieu because not all students are innately resilient and motivated, thus, leveling the field for all nursing students.

Ethics approval number

The Ethics Review Committee (ERC) of Universidad de Manila (Protocol No.: UdM-ERC-0024, approved: 05/17/2022) granted permission to conduct the study.

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CRediT authorship contribution statement

DJEB was involved in the development of the survey questionnaire. Made substantial contributions to conception and design, data acquisition, and data analysis and interpretation. Involved in drafting the manuscript or revising it critically for important intellectual content. Given final approval of the version to be published. The author has participated sufficiently in the work to take public responsibility for the content. Agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data sharing statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nepr.2023.103639.

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